

(I)

Scatt. Theory

(II)

Nuclear
Phys.

(III)

Particle
Phys.

(I)

Scattering Theory

$$\sigma = \int d\Omega \frac{d\sigma}{d\Omega} \quad \underbrace{j}_{\substack{\uparrow \\ \text{incoming flux}}} \underbrace{b db d\phi}_{d\sigma} = \underbrace{n}_{\substack{\uparrow \\ \# \text{ particles} \\ \text{per solid angle}}} \underbrace{\sin\theta d\theta d\phi}_{d\Omega}$$

$$\frac{d\sigma}{d\Omega} = \frac{n}{j} = \frac{b}{\sin\theta} \left| \frac{db}{d\theta} \right|$$

① Draw a diagram for illustration.

② find $b = b(\theta)$ explicitly

(using angular momentum conservation
and Newton's 2nd law $dp = \int F dt$)

Classical

Rutherford scattering $\rightarrow \frac{d\sigma}{d\Omega} \propto \frac{1}{\sin^4(\frac{\theta}{2})}$

Quantum (SI8-related)

$$\frac{d\sigma}{d\Omega} = |f(\underline{k}, \underline{k}')|^2$$

Form factor:

$$F(\underline{q}) = \frac{1}{Z} \int d^3\underline{x} \rho(\underline{x}) e^{i\underline{q} \cdot \underline{x}}$$

$$\frac{d\sigma}{d\Omega} = |F(\underline{q})|^2 \left(\frac{d\sigma}{d\Omega} \right)_{\text{Rutherford (point-charge)}}$$

Fourier transform of charge dist.,
conveying info. about structure.

Born Approx. (1st order P.T.)

$$f(\underline{k}, \underline{k}') = -\frac{m}{2\pi\hbar^2} \int d^3\underline{r} e^{i\underline{q} \cdot \underline{r}} \underbrace{\Delta H(\underline{r})}_{V(\underline{r})}$$

Propagator $G = \frac{1}{\frac{\hbar^2 k^2}{2m} - E}$ (virtual particle)

Fermi's Golden Rule

$$R = \frac{2\pi}{\hbar} \underbrace{|\langle f | \Delta H | i \rangle|^2}_{M_{fi}^2} \rho(E)$$

Resonant Scattering

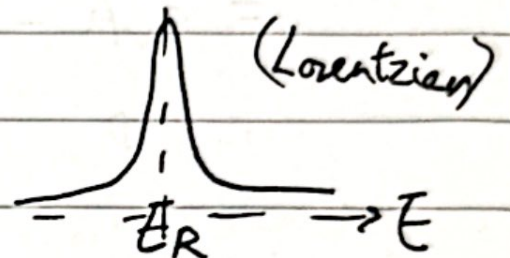
$$\Delta E \text{ st } \sim \hbar \quad \Delta E \sim \hbar \Gamma$$

small $\Gamma \rightarrow$ long-lived states $\rightarrow$ 'particles' (well-defined E)

large $\Gamma \rightarrow$ short-lived states $\rightarrow$ transitional state.

Breit-Wigner

$$\sigma_E = \frac{\pi}{k_i^2} \frac{\Gamma_i \Gamma_f}{(E_R - E)^2 + \frac{\Gamma^2}{4}}$$



Γ : full-width at half-max $\Gamma = \sum_j \Gamma_j$
Sum of all possible decays.

Γ_i : partial width of resonant decay to the initial state.

Γ_f : partial width of resonant decay to the final state.

E_R : resonance energy

k_i : wave number of incoming particles

Branching ratio $B_j = \frac{\Gamma_j}{\Gamma}$

② Nuclear Physics

• SEMF

$\max B(A, Z) \Leftrightarrow \text{stability}$

$$Mc^2 = Zm_p c^2 + (A-Z)m_n c^2 - B(A, Z)$$

where $B(A, Z)$

$$= \underbrace{a_v A}_{(1)} - \underbrace{a_s A^{2/3}}_{(2)} - \underbrace{a_a \frac{(N-Z)^2}{A}}_{(3)} - \underbrace{a_c \frac{Z^2}{A^{1/3}}}_{(4)} \pm \underbrace{\delta(N, Z)}_{(5)}$$

{ (1), (2) : Volume term & surface term:

liquid-drop model

$$E = -C_1 N + 4\pi R^2 T$$

particles in the drop

$$B = -E$$
$$N \sim A \text{ and } R \propto N^{1/3} \sim A^{1/3}$$

(3): Asymmetry term: An imbalance of the occupied states will increase the total energy of the nucleus.
(Fermi statistics) (favors $N=Z$)

④: Coulomb term: $E \sim \frac{Z^2}{r} \sim \frac{Z^2}{A^{1/3}}$
 (repulsion within nucleus)
 favours neutral nuclei

⑤: Pairing term: due to spin states of nucleons.

Even-even	(N-Z)	have	+ δ	↑ more tightly bound
Even-odd	(N-Z)	have	0	
odd-odd	(N-Z)	have	- δ	

Consider ③ & ④ (Z-dependent)

$$\left(\frac{\partial B}{\partial Z}\right)_A = -a_A \frac{2(A-2Z)(-2)}{A} - a_c \frac{2Z}{A^{1/3}} = 0$$

$$+4a_A - 8\frac{a_A Z}{A} = a_c \frac{2Z}{A^{1/3}}$$

$$4a_A(N+Z) - 8a_A Z = 2a_c Z A^{2/3}$$

$$4a_A N - 4a_A Z = 2a_c Z A^{2/3}$$

(valley
of stability)

$$\frac{N}{Z} = 1 + \frac{a_c}{2a_A} A^{2/3} > 1$$

• More neutron rich at large A.

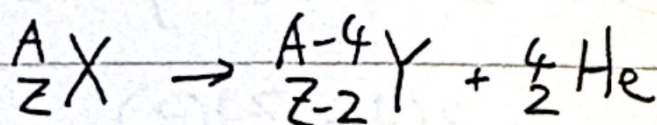
Q-value

$$Q = \sum_i M_i c^2 - \sum_f M_f c^2$$

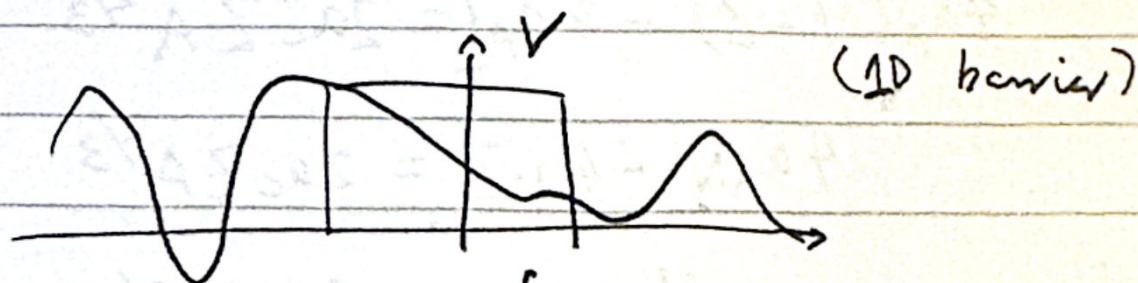
$\Uparrow$ $\Uparrow$
initial final

- larger Q means more likely to occur.
- $Q > 0$ for favourable reactions.

Alpha Decay



- To have the process occur, one must tunnel through the Coulomb and nuclear potential barrier.



$$\psi \propto e^{-\int k dr}$$

$$k = \sqrt{\frac{2m}{\hbar^2}(V_0 - E)}$$

$$P_{\text{tunnelling}} \propto e^{-2\int k(r) dr} = e^{-2G}$$

$$G = \int k(r) dr = \sqrt{\frac{2m}{\hbar^2}} \int \sqrt{V(r) - Q} dr$$

$$V(r) = \frac{2(Z-2)e^2}{4\pi\epsilon_0 r}$$

$$Z_1 = 2$$

$$Z_2 = Z - 2$$

if $Q \ll V$,

$$G \approx \sqrt{\frac{2m}{\hbar^2}} \sqrt{2(Z-2)\hbar c \alpha_{EM}} \int_{R_a}^{R_b} r^{-\frac{1}{2}} dr$$

$$(R_a \ll R_b) \approx \frac{(Z-2)e^2}{\pi\epsilon_0} \sqrt{\frac{2m}{Q}} \propto \frac{Z-2}{\sqrt{Q}}$$

$$R_a \approx r_0 A^{1/3}$$

$\Rightarrow$ size of nucleus

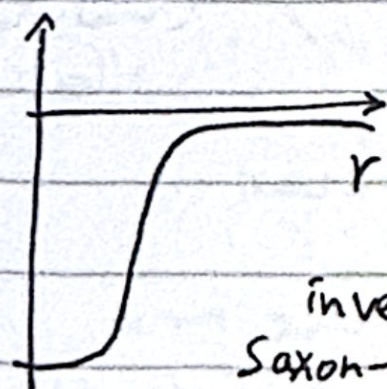
$$R_b = \frac{2(Z-2)\hbar c \alpha_{EM}}{Q}$$

$\Rightarrow$ radius where
KE = PE $\rightarrow$ 'free'

$$\text{Decay rates } \Gamma = \Gamma_0 e^{-2G}$$

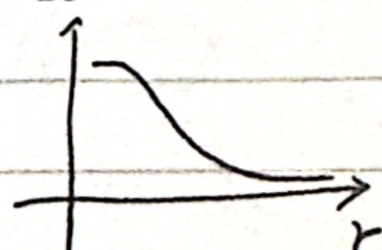
$$\Rightarrow \ln \Gamma = A - B \frac{(Z-2)}{\sqrt{Q}}$$

V_{nuclear}



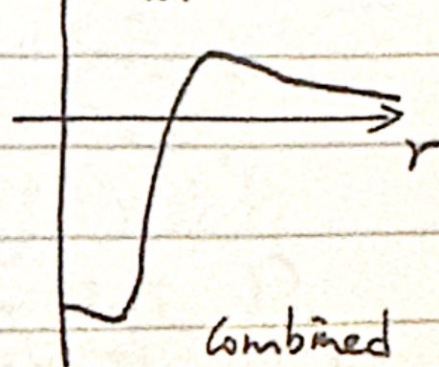
inverted
Saxon-woods

V_{ele}



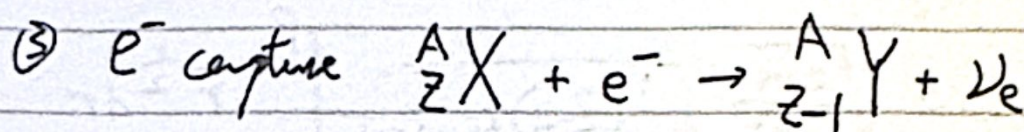
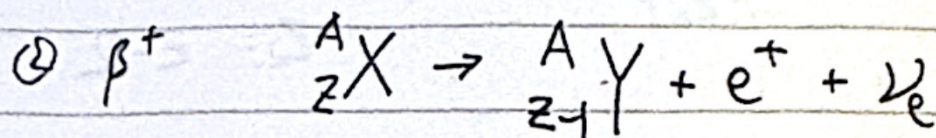
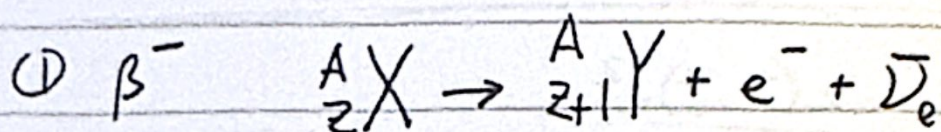
Coulomb

V_{tot}



Combined

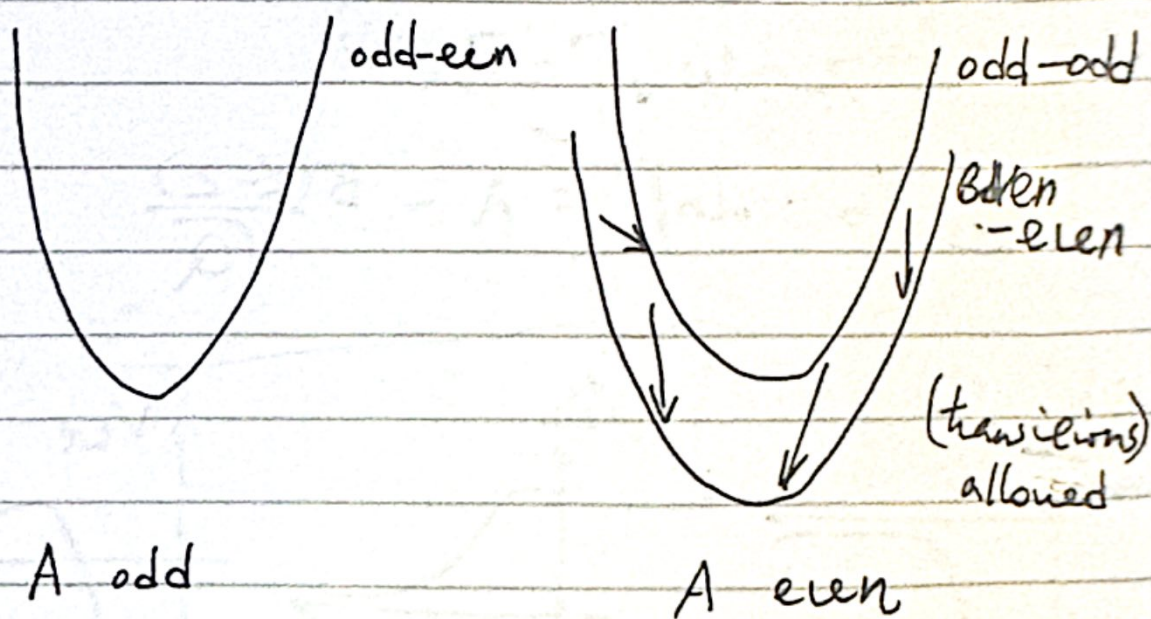
Weak Decays



• only ① has $Q > 0$, can occur in isolation.

• others occur inside the nucleus.

• A unchanged $\rightarrow$ parent & daughter are isobars



(1 isobar stable)

Can have more than 1 stable isobars

Fermi Theory of Beta Decays

- single vertex with G_F (Fermi coupling constant)
- interaction is spin-independent.
- Ignore Coulomb interactions

$$\phi_e = e^{i\mathbf{p}_e \cdot \mathbf{x} / \hbar} \quad \phi_\nu = e^{i\mathbf{p}_\nu \cdot \mathbf{x} / \hbar}$$

(plane-wave solns for states)

- Nucleus too massive,
so just have e^- & $\bar{\nu}$ continuing
to energies: $Q = E_e + E_\nu$
(after)

$$\begin{array}{ccccccc} |\Psi_i\rangle = |\psi_i\rangle & & |\Psi_f\rangle = |\psi_f\rangle |\phi_e\rangle |\phi_\nu\rangle \\ \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \uparrow \\ \text{initial state} & \text{initial nucleus} & \text{final state} & \text{final nucleus} & e^- & \bar{\nu} \end{array}$$

so

$$M_{fi} = \langle \Psi_f | H | \Psi_i \rangle = G_F \int d^3x \phi_e^* \phi_\nu^* \psi_f^* \psi_i$$

$$(\mathbf{p} \cdot \mathbf{x} \ll 1)$$

$$\approx G_F \int d^3x \psi_f^* \psi_i$$

$$\equiv G_F M_{\text{nuc.}}$$

$|M_{nnd}| \approx 1$ ψ_f & ψ_i occupy same states
for an isolated $n \rightarrow p$

POS:

$$d^2N = \frac{d^3p_e}{(2\pi\hbar)^3} \frac{d^3p_\nu}{(2\pi\hbar)^3} = \frac{p_e^2 p_\nu^2}{4\pi^4 \hbar^6} dp_e dp_\nu$$

Consider $\bar{\nu}_e$ basically 'massless',

$$dN = \frac{p_e^2}{4\pi^4 \hbar^6} dp_e \frac{1}{c^3} \int dE_\nu E_\nu^2$$

$$= \left(\frac{p_e^2 (Q - E_e)^2}{4\pi^4 \hbar^6 c^3} dp_e \right) dE_\nu$$

transition
rate:

$$\Gamma(p_e) dp_e = \frac{2\pi}{\hbar} |M_{fi}|^2 \frac{dN}{dE_\nu}$$

$$= \frac{G_F^2 |M_{nnd}|^2}{2\pi^3 \hbar^7 c^3} \underbrace{F(Z, p_e)}_{\text{Fermi factor}} p_e^2 (Q - E_e)^2 dp_e$$

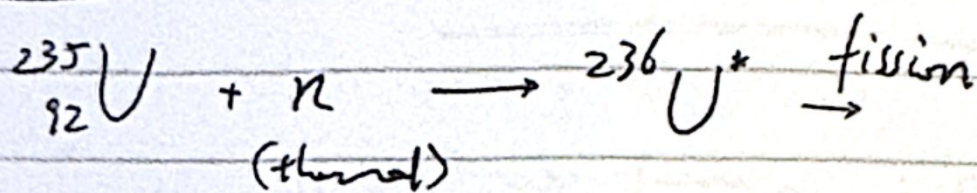
Fermi factor

(correction due
to electromagnetic
interactions between

for ultra-
relativistic limit

$$\Gamma \propto \int_0^Q E_e^2 (Q - E_e)^2 dE_e \propto Q^5 \quad \begin{matrix} e^- \& \text{daughter nucleus} \\ \text{(Sargent's rule)} \end{matrix}$$

Fission



(even-odd)

(even-even)

release of pairing energy

→ sufficient energy to overcome Coulomb barrier.

- Natural U: neutrons are captured by U-238. To achieve chain reactions one needs to (1) enrich U-235 (2) slow down neutrons to thermal energies before captured by U-238.

Fusion

$$P_{\text{fusion}} \propto e^{-\frac{\alpha}{\sqrt{E}}} e^{-\frac{E}{k_B T}}$$

~~~~~  
tunnelling

~~~~~  
Boltzmann factor

(high-energy tail)

$E_{\text{max at}} \quad E \approx \left(\frac{\alpha k_B T}{2} \right)^{2/3}$

(best to overcome Coulomb)

(III) Particle Physics

Isospin: internal quantum # to
tell about the quark content.

$|I, I_3\rangle$ convention $(2I+1 \text{ states for a given multiplet } I)$
 $p = |\frac{1}{2}, \frac{1}{2}\rangle, n = |\frac{1}{2}, -\frac{1}{2}\rangle$
(nucleon doublet)

$$\pi^+ = |1, 1\rangle \quad \pi^0 = |1, 0\rangle \quad \pi^- = |1, -1\rangle$$

$$\Delta^{++} = |\frac{3}{2}, \frac{3}{2}\rangle \quad \Delta^+ = |\frac{3}{2}, \frac{1}{2}\rangle$$

$$\Delta^0 = |\frac{3}{2}, -\frac{1}{2}\rangle \quad \Delta^- = |\frac{3}{2}, -\frac{3}{2}\rangle$$

Isospin is conserved in strong int.

Parity: reversal operation:

$$P: \underline{x} \rightarrow -\underline{x}$$

Apply parity inversion to a state of
well-defined orbital angular momentum:

$$P|n, l, m_l\rangle = (-1)^l |n, l, m_l\rangle$$

$$P_{ab} = P_a P_b (-1)^L \quad \nwarrow \text{total } L = \sum_i l_i$$

↑
two-particle system

- conserved in non-weak interactions

Charge Conjugation

$$C: a \rightarrow \bar{a} \quad \text{particle} \rightarrow \text{antiparticle}$$

For a system of quantum # J, L, S

$$C|a, \bar{a}, J, L, S\rangle = (-1)^{L+S} |a, \bar{a}, J, L, S\rangle$$

Convention:
(subatomic particle)

$$A \overset{\text{charge}}{\overset{Q}{\leftarrow}} J \overset{\text{parity } (\pm)}{\leftarrow} P(C) \overset{\text{charge conjugation } (\pm)}{\leftarrow}$$

↑ ↑
Name Angular momentum

Mesons

$q\bar{q}$, integer spins
 $\uparrow \quad \nwarrow$
 $P=1 \quad P=-1 \quad P_{\text{meson}} = (-1)^{L+1}$

(u & d quarks)

	Quarks	Q	I	I_3	S	B	J
π^+	$u\bar{d}$	1	1	1	0	0	0
π^0	$u\bar{u}, d\bar{d}$	0	1	0	0	0	0
π^-	$d\bar{u}$	-1	1	-1	0	0	0

$$m_{\pi^\pm} \approx 139.6 \text{ MeV}c^{-2}$$

$$m_{\pi^0} \approx 135.0 \text{ MeV}c^{-2}$$

Fid (penciles) $m \sim 500 \text{ MeV}c^{-2}$

(mesons) $m \sim 1200 \text{ MeV}c^{-2}$

decay on timescales of weak decays
 $(\sim 10^{-12} \text{ s})$

$\rightarrow$ strange quarks (s)

s has strangeness $S=-1$, conserved in
 strong but not weak int.

• $J^P = 0^-$ nonet (9) $L=S=J=0$

• decay weakly ($\sim 10^{-10}s$) (antialigned quarks)
 $\approx$ EM

• $J^P = 1^-$ nonet (9) $L=0, J=S=1$

(aligned quarks)
• decay strongly ($\sim 10^{-24}s$)
into 0^- mesons.

• Same quark composition but very different lifetimes

↳ Meson mass mostly contributed by spin dependence (interactions)

• Quark flavour (u, d, s, etc.) is conserved throughout strong force.

Baryons

3 quarks, half-int. spins

$$P_{\text{baryon}} = (-1)^L$$

$L=0$ states: $JP = \frac{1}{2}^+$ or $JP = \frac{3}{2}^+$

(8 particles)

weak,
EM

(10 particles)

strong,
weak
(exception)

into $JP = \frac{1}{2}^+$

Quark flavour & colour

$$|\psi\rangle = |\psi_{\text{space}}\rangle |\psi_{\text{flavour}}\rangle |\psi_{\text{spin}}\rangle |\psi_{\text{colour}}\rangle$$

to ensure overall
antisymmetry

(i.e. uuu, ddd)

for fermions.

(i.e.
Slater
det.)

Quark-colour evidence

Consider the processes:
$$\begin{cases} e^+ + e^- \rightarrow \mu^+ + \mu^- \\ e^+ + e^- \rightarrow \text{hadrons} \end{cases}$$

$$R = \frac{\sigma(e^+e^- \rightarrow \text{hadrons})}{\sigma(e^+e^- \rightarrow \mu^+\mu^-)} = \frac{3 \times g_{EM}^2 \sum_i Q_i^2}{1 \times g_{EM}^2} = 3 \sum_i Q_i^2$$

- EM vertex couplings
- Q_i relative charge
- 3 $\rightarrow$ quark colours (r, g, b)

Graphic $\rightarrow R = 2, \frac{10}{3}, \frac{11}{3}, \dots$

uds udsc udscb

Quarkonium & Charmonium

Non-rel. model for bound states
(sharp resonances)

$$V(r) = -\frac{4}{3} \frac{k_C \alpha_s}{r} + kr$$

$\xrightarrow{\text{Coulomb-equivalent}}$ quark confinement $r \rightarrow \infty$.

$$\alpha_s \approx 1$$

Bound states (small r) $E_n \propto -\frac{1}{n^2}$

hydrogenic-like

Total E for $q\bar{q}$ states: $E = E_n + 2m_q c^2$

$(c\bar{c})$

Charmonium low lying states:

$\eta_c, 3/4, h_c, \psi', \text{ etc}$

i.e. $e^+ + e^- \rightarrow \gamma^* \rightarrow J/\psi$

Possible decay paths:

① If $m_{c\bar{c}} > 2m_D$, strong decay into D-mesons ($c\bar{u}, c\bar{d}, \bar{c}u, \bar{c}d$) (pair)

② $m_{c\bar{c}} < 2m_D$, EM decay through annihilation and produce γ and $q\bar{q}$ pair.

③ Suppressed strong decay

Resulting state contains no c or $\bar{c}$,

so intermediate states consist of only
 gluons, poor momentum coupling
 (OZI suppression)

($\pi^+ \pi^0 \pi^-$ result
 usually)

decay rate $\downarrow\downarrow$

Thus $m_{cc} < 2m_D$ states are long-lived.
 ($c\bar{c}$)

$$\alpha_s > \alpha_w > \alpha_{EM}$$

$$\tau_{strong} \ll \tau_{EM} \ll \tau_{weak}$$

$$(\sim 10^{-22} s) \quad (\sim 10^{-18} s) \quad (\sim 10^{-10} s)$$

Leptons

$$\Gamma \propto Q^5 \propto m_l^5$$

(Sargent's rule,
 ultra-relativistic)

$$\Gamma \propto g^2 m^5 \propto T^{-1}$$

Lepton universality ($g_\mu \approx g_\tau$)

$$\text{so } \underbrace{\frac{B_\tau}{B_\mu}}_{\approx 0.178} \frac{\tau_\mu}{\tau_\tau} \left(\frac{m_\mu}{m_\tau} \right)^5 = \left(\frac{g_\tau}{g_\mu} \right)^2 \approx 1$$

I II III (generations)

$$\begin{pmatrix} u \\ d \end{pmatrix} \begin{pmatrix} c \\ s \end{pmatrix} \begin{pmatrix} t \\ b \end{pmatrix}$$

Cross-generations transitions are suppressed ($\sin\theta_c$), while same generation dominate ($\cos\theta_c$).

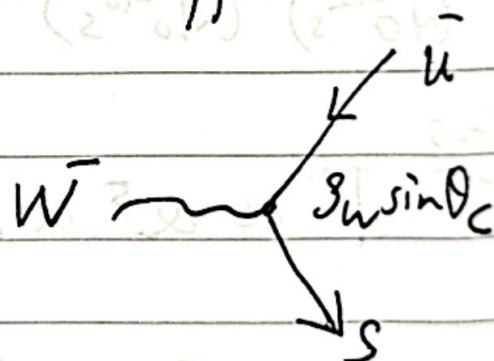
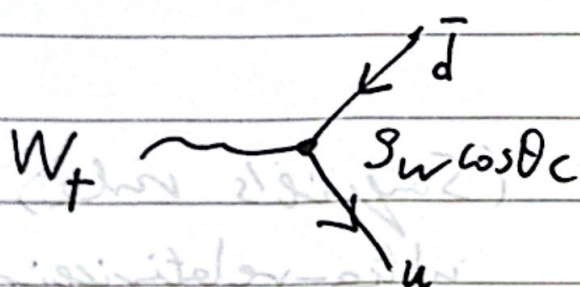
~~Cabibbo~~ Cabibbo angle $\theta_c \approx 13^\circ$

2x2 Cabibbo rotation matrix

3x3

CKM

(I & III most suppressed)



Further Notes

propagator factor $\frac{1}{Q^2 - m^2}$

$$g_{EM} = \sqrt{4\pi\alpha_{EM}}$$

$$\sigma \propto \frac{\alpha_{EM}^2}{s} \quad \text{for s-channel}$$

where $s = E_e^2$

$e^+e^- \rightarrow \mu^+\mu^-$
process.

Saxon-Woods potential / dichiein

$$p(r) = \frac{p_0}{(1 + e^{(r-a)/d})}$$

Cannon factor $\int_{r_a}^{r_b} \rightarrow \frac{2(Z-2)e^2}{4\pi\epsilon_0 Q}$
 $r_a \rightarrow r_0 A^{1/3}$

Use $r = r_b \cos^2\theta$ substitution.

Consider $\frac{\hbar^2 k^2}{2m} = \cancel{V(r)} V(r) - Q.$

$\pi^0 \rightarrow \gamma\gamma\gamma$ me allowed

$$(1) \uparrow C_{\pi^0}$$

$$(-1)^3 \uparrow C_\gamma$$

(charge conjugation
not concerned)

quarks pulled away $\rightarrow$ form new hadrons

2-jets: from hadronisation of gluons

3-jets: one gluon ejected earlier and then hadronises

↓
bundle of particles within a narrow cone.

$\alpha_{EM} \uparrow$ with $E \uparrow$ but $\alpha_s \downarrow$
(slightly) (strongly)

Q values for β^- : $m_p - m_D$
 β^+ : $m_p - m_D - 2m_e$
EC: $m_p - m_D$